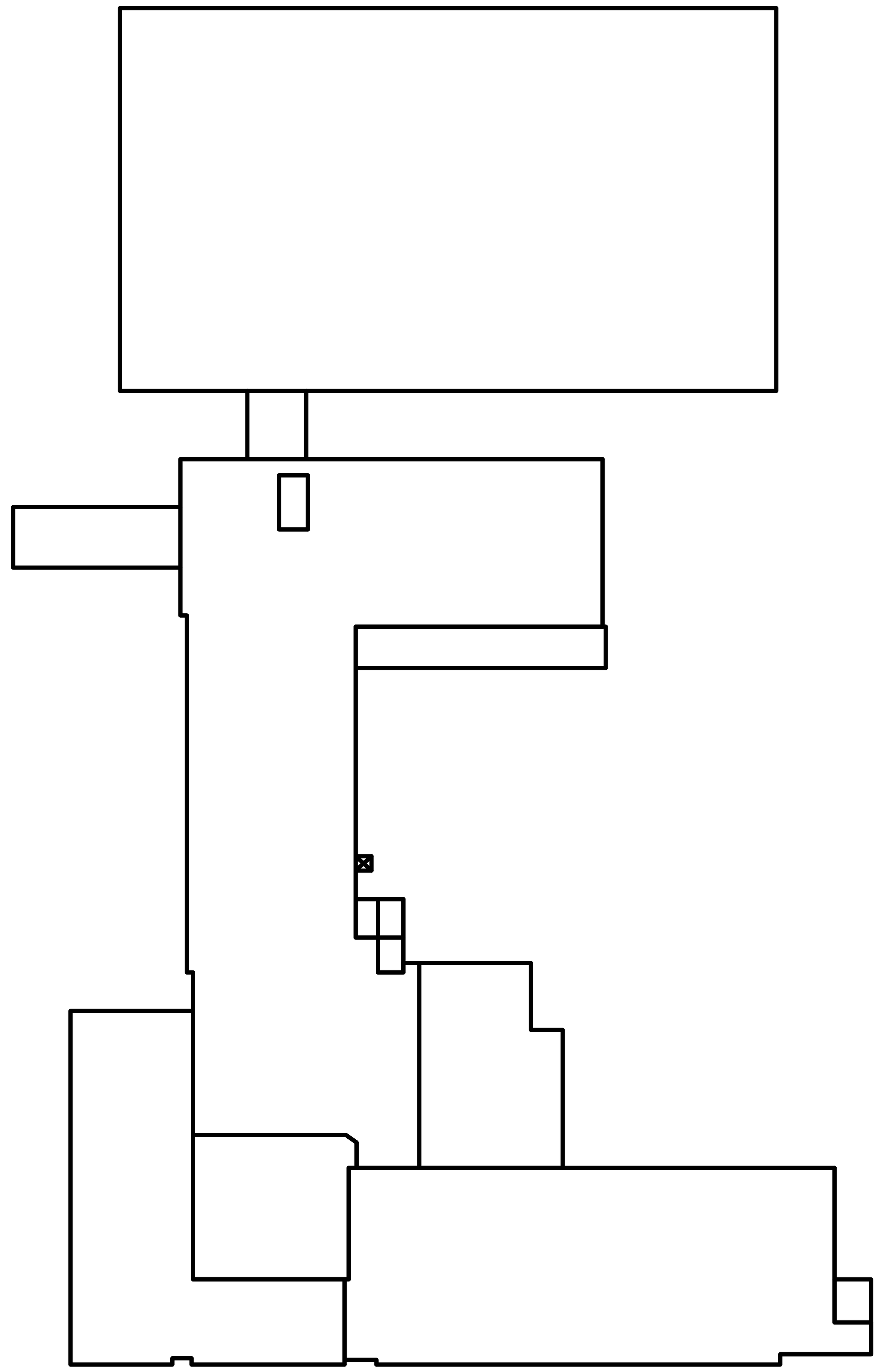
ROOF DIAGNOSTIC SURVEY FOR UNITED STATES AIR FORCE

SHAW AFB BUILDING 1130

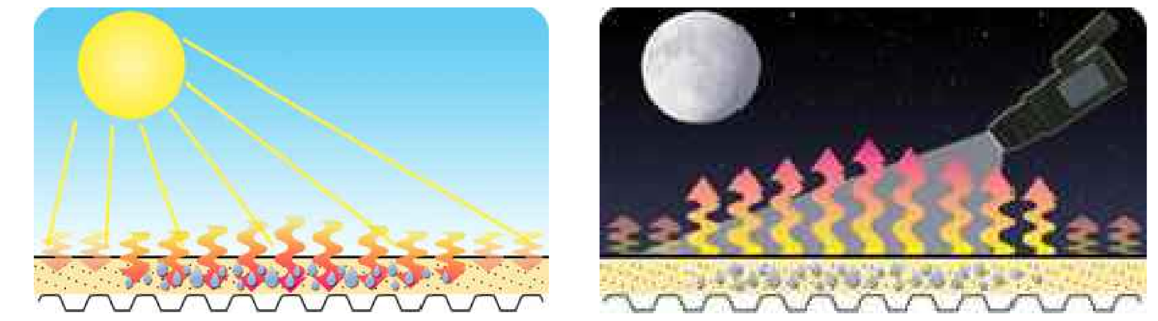
420 CHAPIN STREET SHAW AFB, SOUTH CAROLINA 29152

DRAWINGS



ROOF PLAN
SCALE: NO SCALE

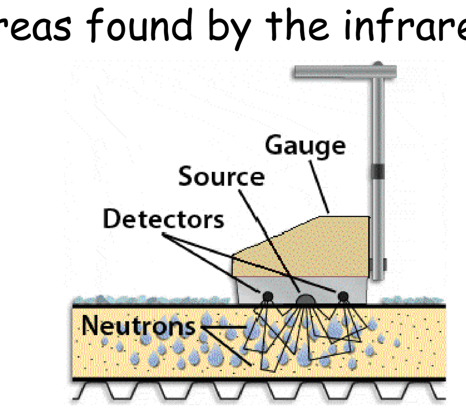
How An Infrared Survey Works:



Wet roof insulation absorbs more solar energy from the sun during the day than dry insulation. As the sun sets, the nighttime cools the roof surface. Wet insulation retains more solar energy than dry insulation. These temperature differences are detected by the infrared camera. All full service scans are performed using a mid-wave IR camera with a 2.5 - 5.6 spectral range. Suspected wet insulation is verified through core cuts and/or a Roof Moisture Meter. Core cuts are taken to establish acceptable levels of moisture. Confirmed wet areas are marked out on the roof surface with paint.

How A Moisture Meter Works:

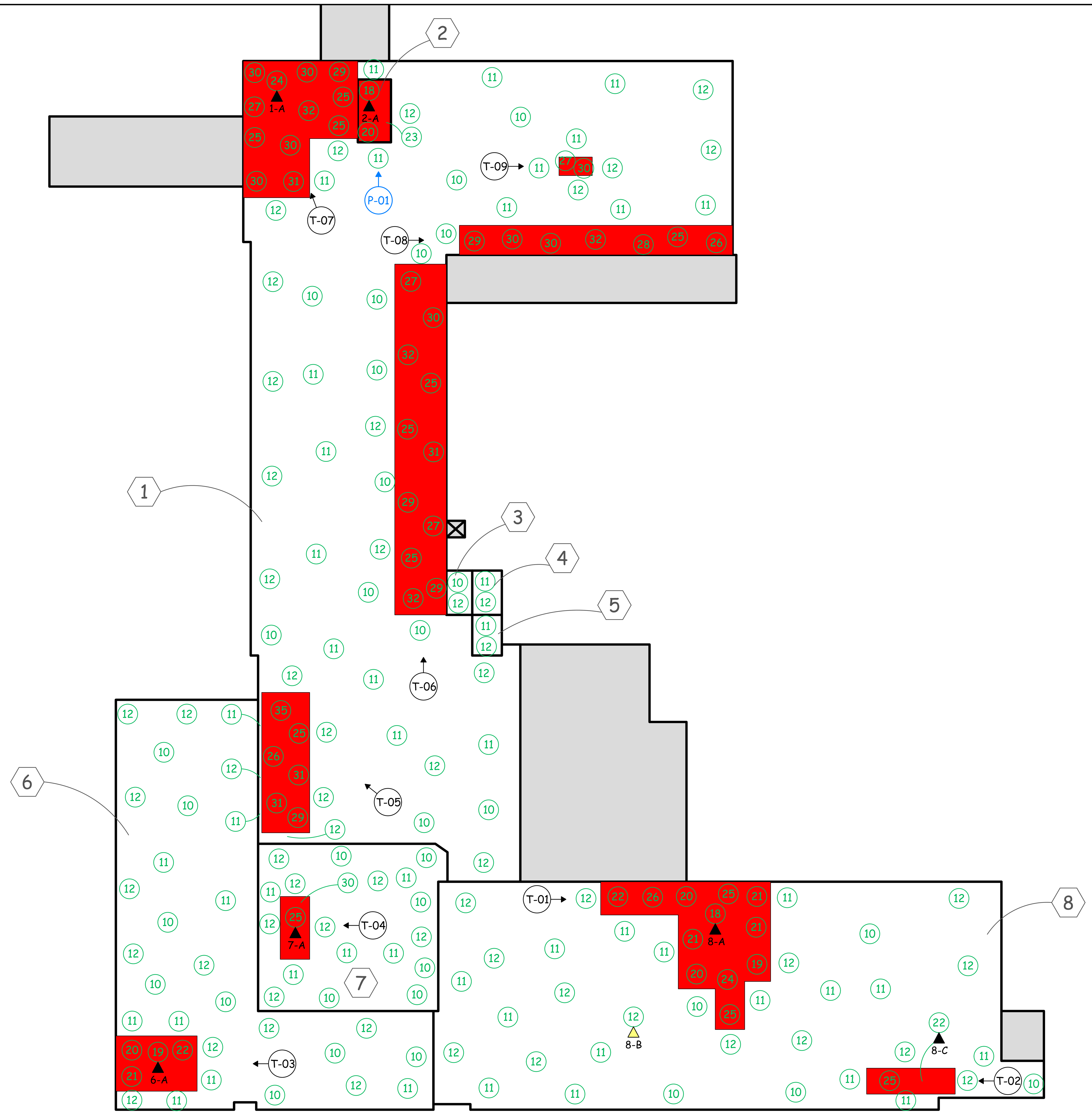
During the daytime, readings are taken and recorded in random locations and at wet areas found by the infrared camera.



Fast neutrons are emitted from the source in the Roof Moisture Meter into the roof system. The presence of hydrogen in the roof system slows the neutrons. These slowed neutrons as well as the fast neutrons are detected by the Roof Moisture Meter. A reading is displayed in the digital readout and gets recorded. Core cuts are taken to determine a baseline for dry roof materials. Wet areas of insulation are marked on the roof surface with paint.

TREMCO Roofing & Building Maintenance

TREMCO'S INFRARED INSPECTIONS ARE PERFORMED BY CERTIFIED (EXPERIENCED) INFRARED THERMOGRAPHERS UTILIZING STATE-OF-THE-ART EQUIPMENT. ALL WORK CONFORMS WITH CURRENT INDUSTRY BEST PRACTICES, AND IS PERFORMED IN ACCORDANCE WITH STANDARDS PUBLISHED BY ASTM INTERNATIONAL AND INFRASPECTION INSTITUTE.



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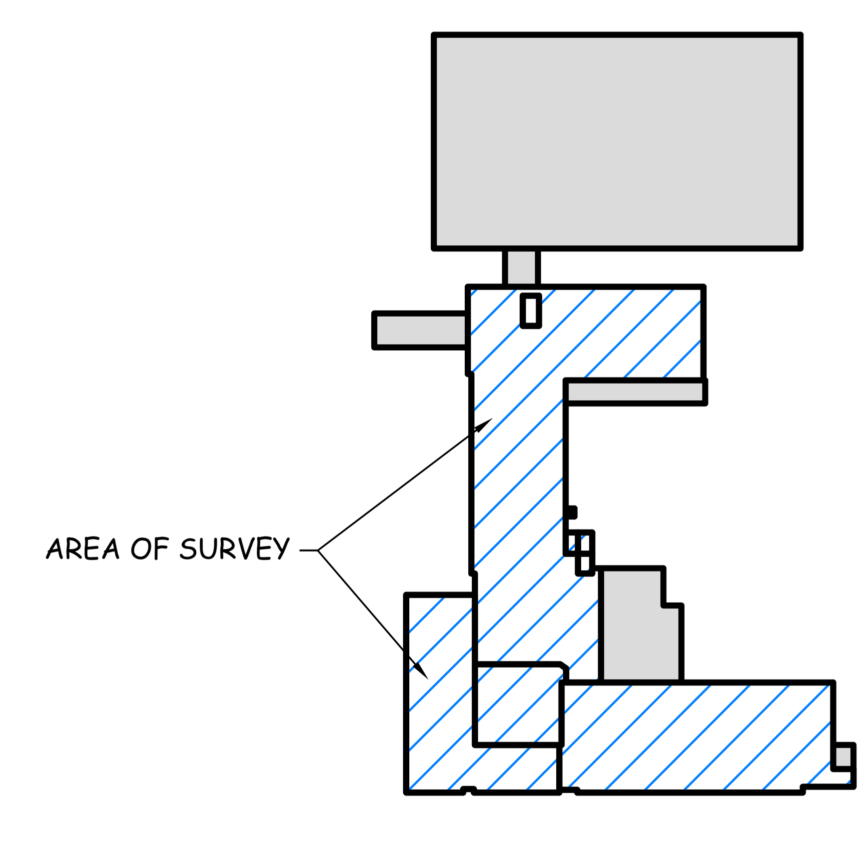

KEY PLAN
SCALE: NO SCALE

STANDARD KEY OF SYMBOLS

# ROOF SECTION	P-# PHOTO		N.I.C.
DRY CORE	T-# THERMOGRAM	WET INSULATION	MOISTURE GRID
WET CORE	N.A.F. (NO ANOMALIES FOUND)	R.I.M. (RANDOM INTERMITTENT MOISTURE)	TRACE CORE
			# MOISTURE READING

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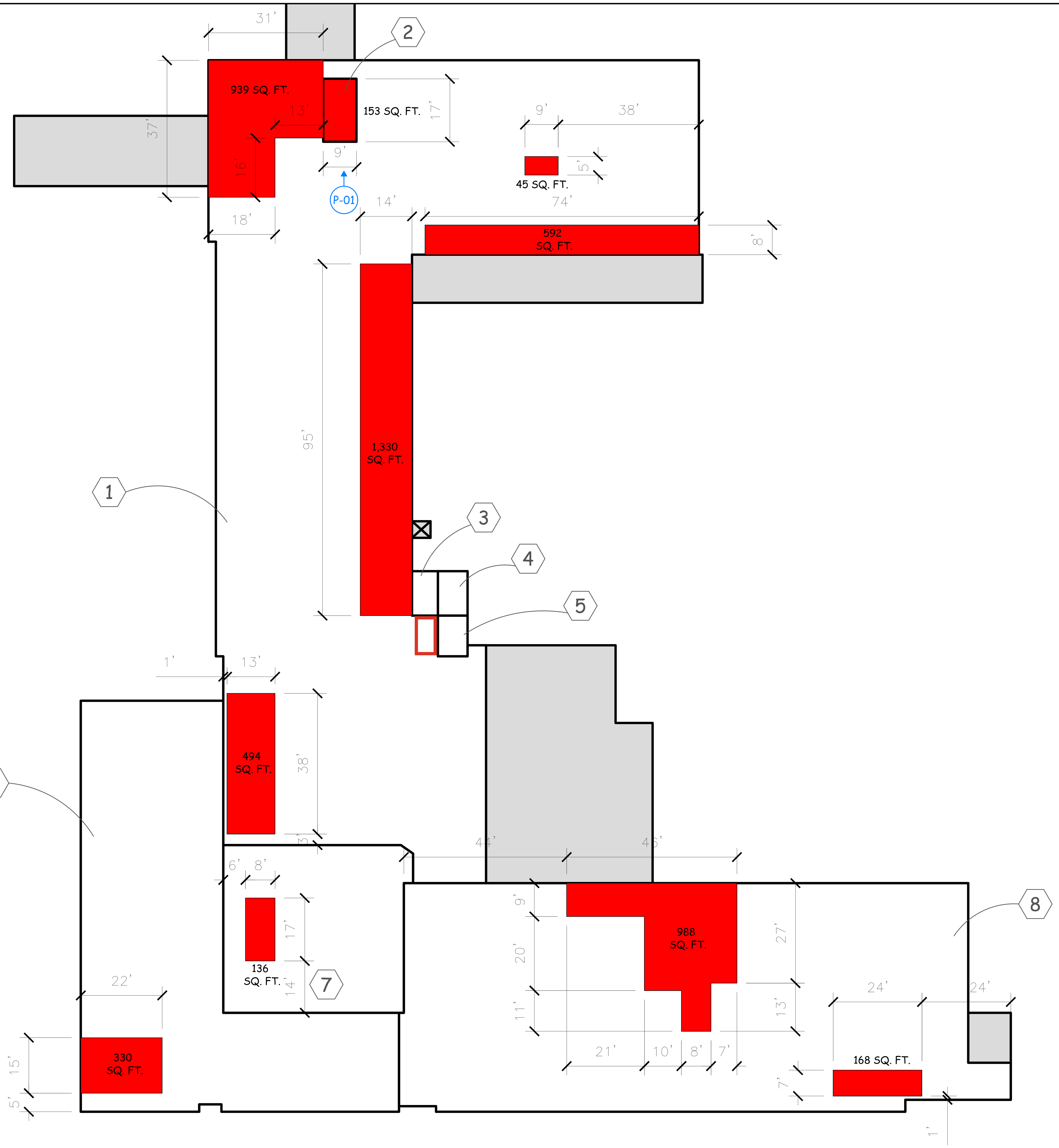
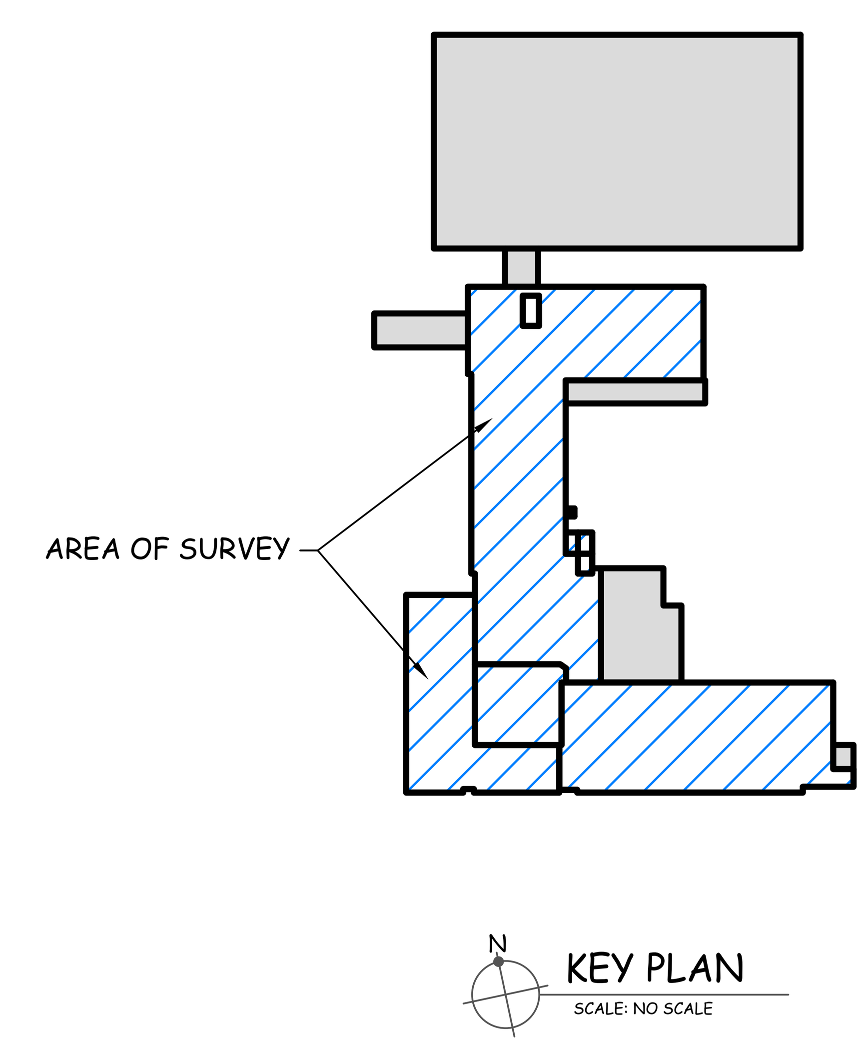
UNITED STATES AIR FORCE - SHAW AFB
BUILDING 1130
420 CHAPIN STREET
SHAW AFB, SOUTH CAROLINA 29152

DIAGNOSTIC TECHNICIAN: M. Sporcic	PROJECT NO.: 9851183
DRAWN BY: R.L.	SHEET NO.: A
DATE: 12/23/2024	



ROOF PLAN
SCALE: NO SCALE



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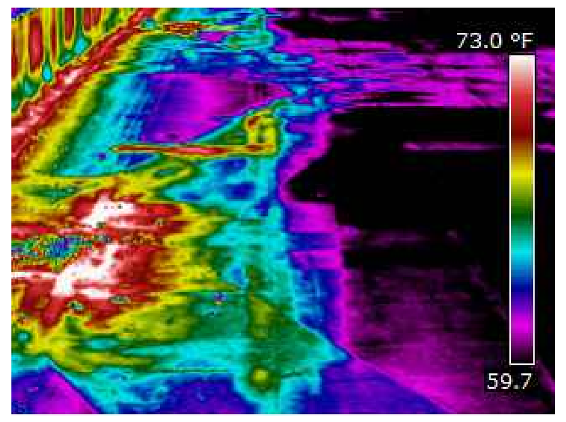
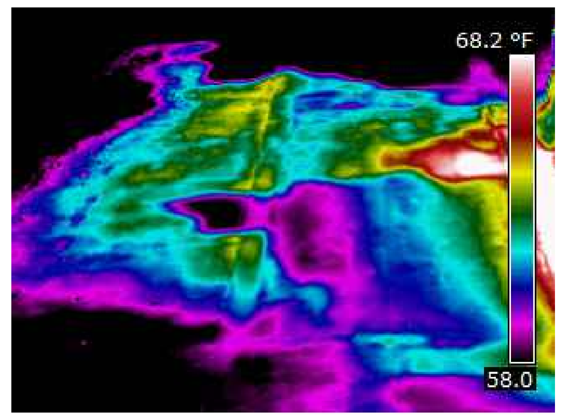
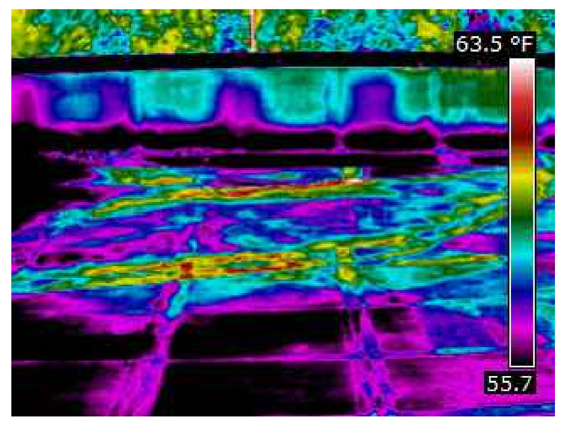
DIAGNOSTIC TECHNICIAN: M. Sporcic	PROJECT NO.: 9851183
DRAWN BY: R.L.	SHEET NO.: B
DATE: 12/23/2024	



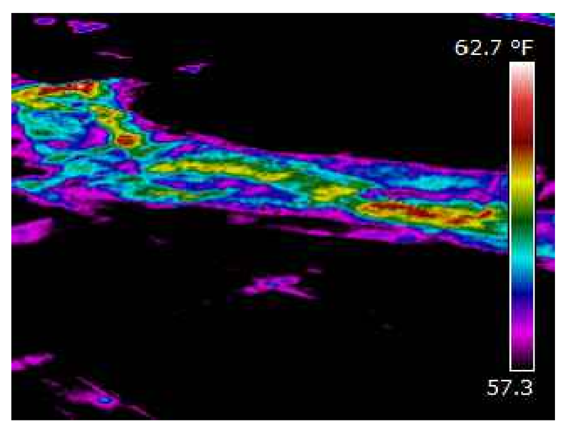
Thermogram T-01



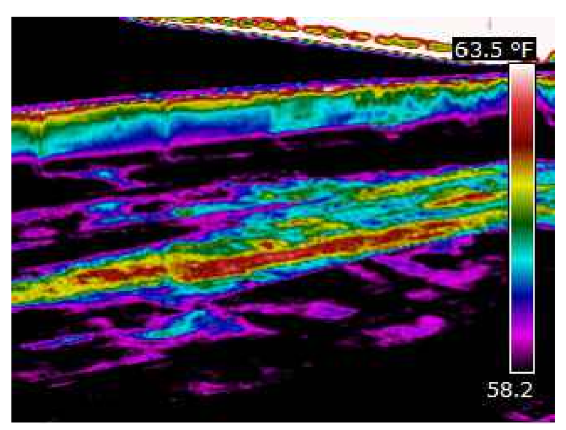
Thermogram T-02



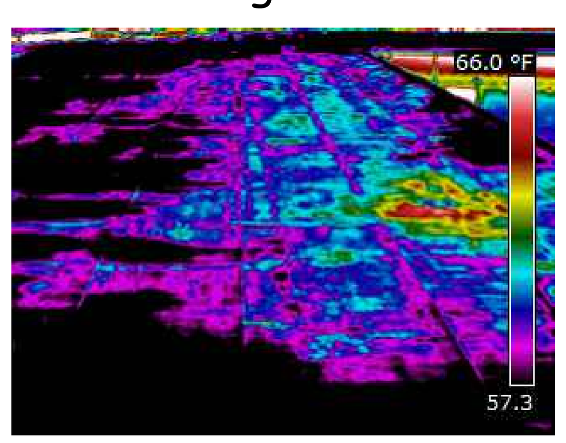
Thermogram T-03



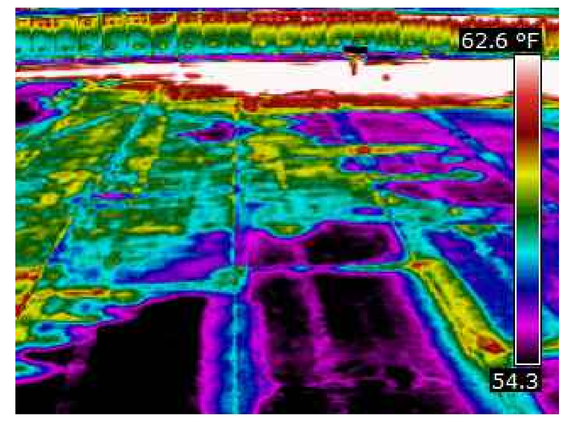
Thermogram T-04



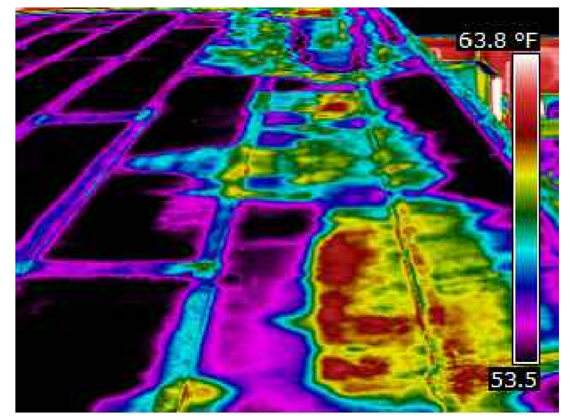
Thermogram T-05



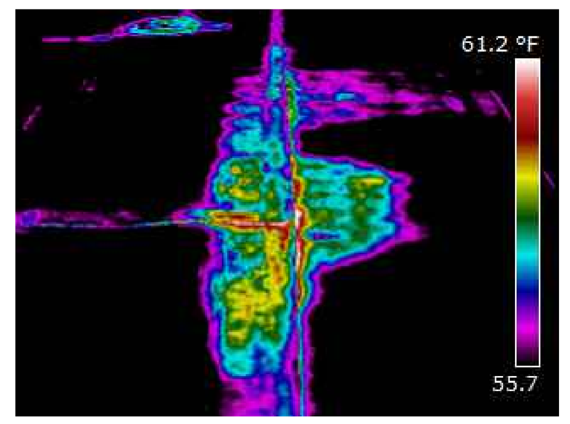
Thermogram T-06



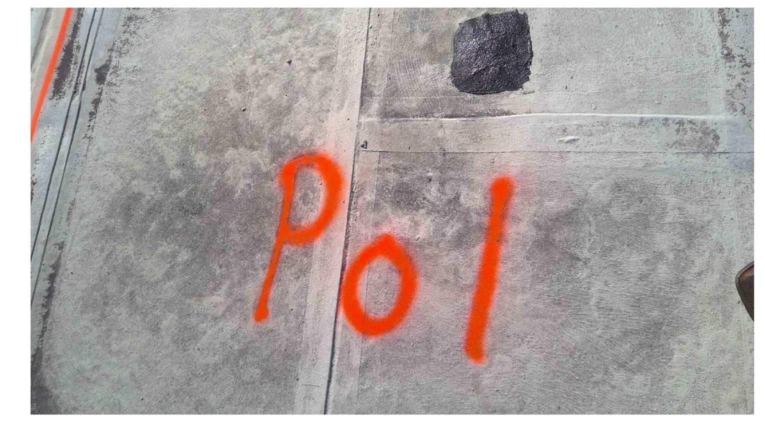
Thermogram T-07



Thermogram T-08



Thermogram T-09



Photograph P-01

ROOF SECTION DATA

ROOF SECTION	SIZE (S.F.)	WET (S.F.)	% WET
1	16,618	3,400	20.46%
2	153	153	100.00%
3	84	0	0.00%
4	96	0	0.00%
5	88	0	0.00%
6	5,531	330	5.97%
7	2,230	136	6.10%
8	9,492	1,156	12.18%
TOTALS	**34,292**	**5,175**	**15.09%**

CONSTRUCTION DATA

ROOF SECTION	CORE CUT NUMBER	MOISTURE READING	MOISTURE PERCENTAGE	ROOF CONSTRUCTION	
1	1-A	24	0%		MODIFIED ROOF SYSTEM W/ GRANULES
			100%	1/2"	DENS DECK
			100%	3-1/2"	POLYISOCYANURATE INSULATION
			0%		CONCRETE DECK
ROOF SECTION	CORE CUT NUMBER	MOISTURE READING	MOISTURE PERCENTAGE	ROOF CONSTRUCTION	
2	2-A	18	0%		MODIFIED ROOF SYSTEM W/ GRANULES
			100%	1/2"	DENS DECK
			100%	3"	POLYISOCYANURATE INSULATION
			0%		CONCRETE DECK
ROOF SECTION	CORE CUT NUMBER	MOISTURE READING	MOISTURE PERCENTAGE	ROOF CONSTRUCTION	
6	6-A	19	0%		MODIFIED ROOF SYSTEM W/ GRANULES
			100%	1/2"	DENS DECK
			100%	1"	POLYISOCYANURATE INSULATION
			0%	3"	POLYISOCYANURATE INSULATION
			0%	3"	POLYISOCYANURATE INSULATION
			0%		CONCRETE DECK
ROOF SECTION	CORE CUT NUMBER	MOISTURE READING	MOISTURE PERCENTAGE	ROOF CONSTRUCTION	
7	7-A	25	0%		MODIFIED ROOF SYSTEM W/ GRANULES
			100%	1/4"	DENS DECK
			100%	1"	POLYISOCYANURATE INSULATION
			100%	2"	POLYISOCYANURATE INSULATION
			100%	1"	POLYISOCYANURATE INSULATION
			50%	1/2"	DENS DECK
			0%		CONCRETE DECK
ROOF SECTION	CORE CUT NUMBER	MOISTURE READING	MOISTURE PERCENTAGE	ROOF CONSTRUCTION	
8	8-A	18	0%		MODIFIED ROOF SYSTEM W/ GRANULES
			100%	1/2"	DENS DECK
			100%	2-1/2"	POLYISOCYANURATE INSULATION
			60%	2-1/2"	POLYISOCYANURATE INSULATION
			0%		CONCRETE DECK
ROOF SECTION	CORE CUT NUMBER	MOISTURE READING	MOISTURE PERCENTAGE	ROOF CONSTRUCTION	
8	8-B	12	0%		MODIFIED ROOF SYSTEM W/ GRANULES
			0%	1/2"	DENS DECK
			0%	2-1/2"	POLYISOCYANURATE INSULATION
			0%	2-1/2"	POLYISOCYANURATE INSULATION
			0%		CONCRETE DECK
ROOF SECTION	CORE CUT NUMBER	MOISTURE READING	MOISTURE PERCENTAGE	ROOF CONSTRUCTION	
8	8-C	22	0%		MODIFIED ROOF SYSTEM W/ GRANULES
			100%	1/2"	DENS DECK
			100%	2-1/2"	POLYISOCYANURATE INSULATION
			100%	2-1/2"	POLYISOCYANURATE INSULATION
			0%		CONCRETE DECK

NOTE: THE MOISTURE PERCENTAGES ARE INTENDED TO BE USED AS A QUALITATIVE MEASUREMENT RATHER THAN AN EXACT READING.

INFRASPECTION INSTITUTE® CERTIFIED Level I Michael Sporcic No: 16139 INFRARED THERMOGRAPHER

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